BASE CHANGE FOR THE DERIVED HOM OUT OF A PERFECT OR PSEUDO-COHERENT COMPLEX IN THE AFFINE CASE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The global hom complex admits a canonical map to the hom complex after base change, and the displayed map of the statement is obtained from it by adjointness. used at proof lines 5447-5455. Produces the comparison map whose bijectivity is asserted.
- For an affine morphism, the pushforward of the pullback of an object is described by tensoring with the pushforward of the structure sheaf. invoked at proof lines 5456-5460. Rewrites the right hand side of the comparison as a derived Hom on the original scheme.
- The derived sheaf Hom out of a perfect or pseudo-coherent complex has the positivity or finiteness properties required, and the internal Hom computation describes it in the tor independent case. invoked at proof lines 5460-5492. Reduces both cases to the comparison statement for the derived sheaf Hom, which is where the two hypotheses enter.

Source reference. Original section 0B6A: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5447-5450 for the canonical map; the adjointness step producing the displayed map is inside the counted span.

Source reference. Original lemma 08I9: sources/stacks-perfect/pinned-original-source.tex, source lines 888–888. Invoked at proof lines 5456-5459 to rewrite the right hand side; the rest of the argument is inside the counted span.

Source reference. Original lemma 0ATN: sources/stacks-perfect/pinned-original-source.tex, source lines 2719–2719. Used at proof lines 5473-5485 to identify both sides and conclude the isomorphism; the reduction to it is inside the counted span.

Source reference. Original lemma 0A6H: sources/stacks-perfect/pinned-original-source.tex, source lines 2657–2657. Cited at proof lines 5465-5472 to see that the derived sheaf Hom lies in the required category under the hypotheses; the use made of that is inside the counted span.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08E9: sources/stacks-perfect/pinned-original-source.tex, source lines 2508–2508. Used at proof lines 5468-5472 to control the second argument using the tor dimension hypothesis; the conclusion is inside the counted span.

Source reference. Original lemma 08IB: sources/stacks-perfect/pinned-original-source.tex, source lines 5362–5362. Cited at proof lines 5486-5492 as the statement that finishes both cases; the reduction carried out inside the counted span is what makes it applicable.

Source reference. Original lemma 09U8: sources/stacks-chow/tags.txt, source lines 1–1. A navigation pointer used while controlling the arguments after base change; the control itself is inside the counted span.

Lemma 0.2. *Consider a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

of quasi-compact and quasi-separated schemes. Assume g and f Tor independent and $S = \mathrm{Spec}(R)$, $S' = \mathrm{Spec}(R')$ affine. For $M, K \in D(\mathcal{O}_X)$ the canonical map

$$R\mathrm{Hom}_X(M, K) \otimes^{\mathbf{L}}_R R' \longrightarrow R\mathrm{Hom}_{X'}(L(g')^*M, L(g')^*K)$$

in $D(R')$ is an isomorphism in the following two cases

1. *$M \in D(\mathcal{O}_X)$ is perfect and $K \in D_{QCoh}(X)$, or*
2. *$M \in D(\mathcal{O}_X)$ is pseudo-coherent, $K \in D^+_{QCoh}(X)$, and R' has finite tor dimension over R.*

Proof. There is a canonical map $R\mathrm{Hom}_X(M, K) \to R\mathrm{Hom}_{X'}(L(g')^*M, L(g')^*K)$ in $D(\Gamma(X, \mathcal{O}_X))$ of global hom complexes, see Cohomology, Section 0B6A. Restricting scalars we can view this as a map in $D(R)$. Then we can use the adjointness of restriction and $- \otimes^{\mathbf{L}}_R R'$ to get the displayed map of the lemma. Having defined the map it suffices to prove it is an isomorphism in the derived category of abelian groups.

The right hand side is equal to

$$R\mathrm{Hom}_X(M, R(g')_*L(g')^*K) = R\mathrm{Hom}_X(M, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} g'_*\mathcal{O}_{X'})$$

by Lemma 08I9. In both cases the complex $R\mathcal{H}om(M, K)$ is an object of $D_{QCoh}(\mathcal{O}_X)$ by Lemma 0A6H. There is a natural map

$$R\mathcal{H}om(M, K) \otimes^{\mathbf{L}}_{\mathcal{O}_X} g'_*\mathcal{O}_{X'} \longrightarrow R\mathcal{H}om(M, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} g'_*\mathcal{O}_{X'})$$

which is an isomorphism in both cases by Lemma 0ATN. To see that this lemma applies in case (2) we note that $g'_*\mathcal{O}_{X'} = Rg'_*\mathcal{O}_{X'} = Lf^*g_*\mathcal{O}_{S'}$ the second equality by Lemma 08IB. Using Lemma 08E9 and Cohomology, Lemma 09U8 we conclude that $g'_*\mathcal{O}_{X'}$ has finite Tor dimension. Hence, in both cases by replacing K by $R\mathcal{H}om(M, K)$ we reduce to proving

$$R\Gamma(X, K) \otimes^{\mathbf{L}}_A A' \longrightarrow R\Gamma(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} g'_*\mathcal{O}_{X'})$$

is an isomorphism. Note that the left hand side is equal to $R\Gamma(X', L(g')^*K)$ by Lemma 08I9. Hence the result follows from Lemma 08IB. □